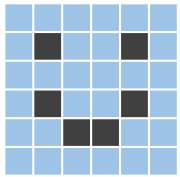
Transformer components

Attention inputs: Q, K, V

A query retrieves relevant information from keys/values
Q (Query): what am I looking for?
K (Key): what information can I offer?
V (Value): the actual information I can retrieve

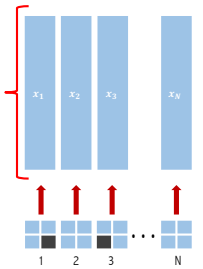


patchify
Or tokenize (text)



N: Input sequence length

Build representations



F: Input embedding dimension

Token matrix X

$X =$ (F)

$X \in \mathbb{R}^{N \times F}$

Project X
To build Q, K, and V

$$Q = X\,W^Q$$
$$K = X\,W^K$$
$$V = X\,W^V$$

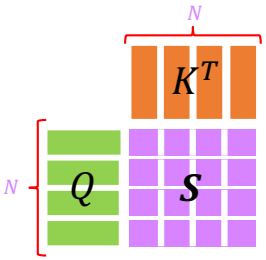
Note: For Cross-attention, Q comes from one source, and K,V from another

X ($N \times F$) W^Q ($F \times D$) $=$ Q ($N \times D$)

- W are learnable parameters; one W matrix per each Q, K, V
- W are indep. from N: in inference it can take different dimension input
- D: Projected Query/Key/Value dimension

Attention computation

$$Scores\ S = \underbrace{Q}_{N \times D} \cdot \underbrace{K^T}_{D \times N} \in \mathbb{R}^{N \times N}$$



- Dot product measures query-key compatibility (alignment).
- Conventional dense self-attention has quadratic complexity $O(N^2)$ with respect to sequence length N.

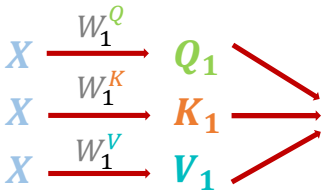
$$Attention\ \boldsymbol{A} = softmax(S/\sqrt{\boldsymbol{D}}) \in \mathbb{R}^{N \times N}$$

- *Softmax* maps raw scores into probability distributions (non-negative values that sum up to one)
- *Softmax* introduces non-linearity
- *Scaling* prevents large dot products from saturating softmax, helping maintain stable gradients.

$$Output\ \boldsymbol{O} = \underbrace{\boldsymbol{A}}_{N \times N}\ \underbrace{\boldsymbol{V}}_{N \times D} \in \mathbb{R}^{N \times D} = softmax\left(\frac{Q \cdot K^T}{\sqrt{\boldsymbol{D}}}\right)\boldsymbol{V}$$

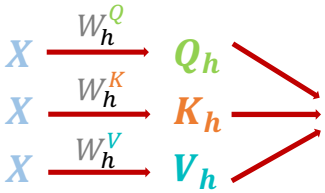
Multi-Head attention

- Head-specific weights
- Each head can focus on a different relationship Q-K, for example:
  In text: grammar, meaning, coreference
  In images: texture, object structures, semantic relationships
- With h heads, the total projection dimension D is typically split across heads



$$\boldsymbol{O_{head1}} = softmax\left(\frac{Q_1 \cdot {K_1}^T}{\sqrt{\boldsymbol{D}}}\right)V_1$$

$$\boldsymbol{O_{MHA}} = Concat(\boldsymbol{O_{head1}}, \ldots, \boldsymbol{O_{headh}})\boldsymbol{W^O}$$



$$\boldsymbol{O_{headh}} = softmax\left(\frac{Q_h \cdot {K_h}^T}{\sqrt{\boldsymbol{D}}}\right)V_h$$

Note: For Cross-attention, input for Q comes from one source, and K,V from X

Positional Embeddings

- Self-attention allows each token to attend to other tokens regardless of their distance, subject to the attention mask. Without positional information, attention cannot distinguish between different arrangements of the of the same tokens.
- Positional information can be added to input embeddings or incorporated directly into attention

$$X \leftarrow X + \boldsymbol{P} \in \mathbb{R}^{N \times F}$$

Additive positional embeddings

Types:
- Absolute:
  Fixed: deterministic mathematical functions (e.g., sinusoidal encodings).
  Learned: let the network let the positional embedding
- Relative: based on relative token positions. e.g. RoPE

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v.1.0